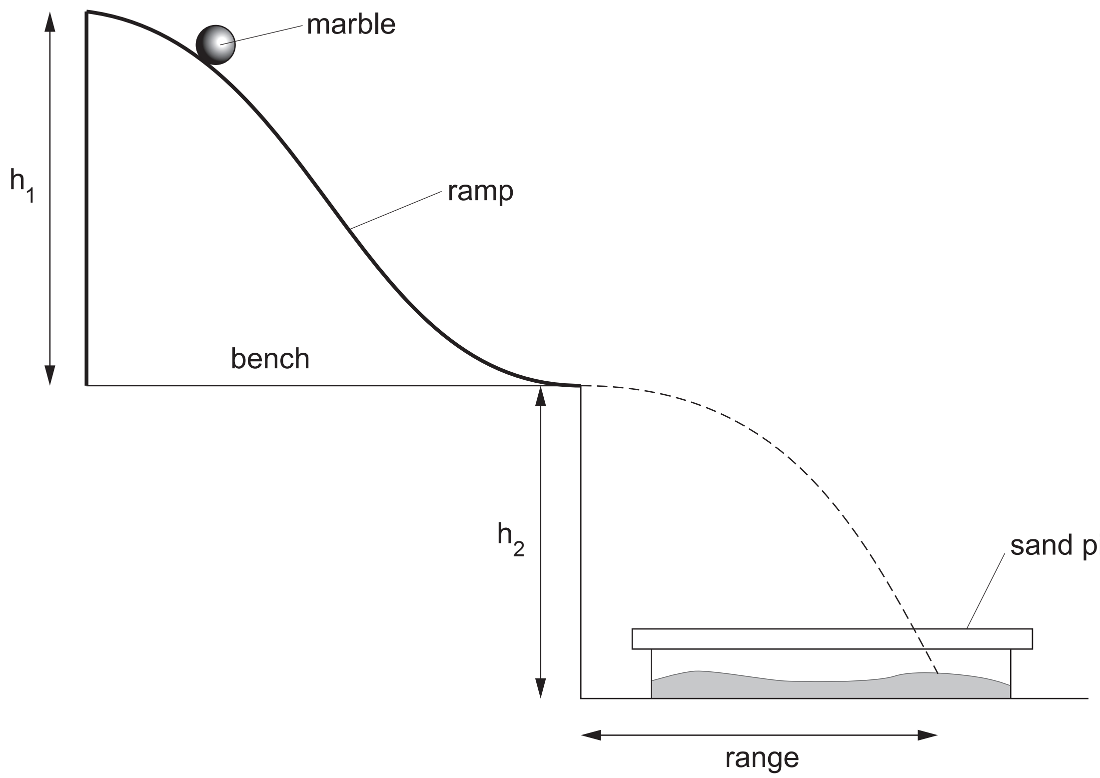
4. A model ski jump can be set up on a bench as shown in the diagram below.



The marble is released from rest at the top of a ramp. It leaves the bottom of the ramp with a certain velocity. It moves through the air and lands in a sand pit. The horizontal distance travelled in the air is called the **range**.

The velocity of the marble as it leaves the ramp is calculated using the equation:

$$\text{velocity} = \frac{\text{range}}{\text{time to travel the range}}$$

The acceleration of the marble down the ramp is calculated using the equation:

$$\text{acceleration} = \frac{\text{velocity of marble as it leaves the ramp}}{\text{time to travel down the ramp}}$$

Examiner only

(a) The following measurements were recorded during one trial of the experiment.

Range = 45 cm
Time to travel the range = 0.5 s
Time to travel down the ramp = 1.2 s

Use the equations and information above to calculate the acceleration of the marble down the ramp in m/s^2. [4]

Acceleration = ... m/s^2

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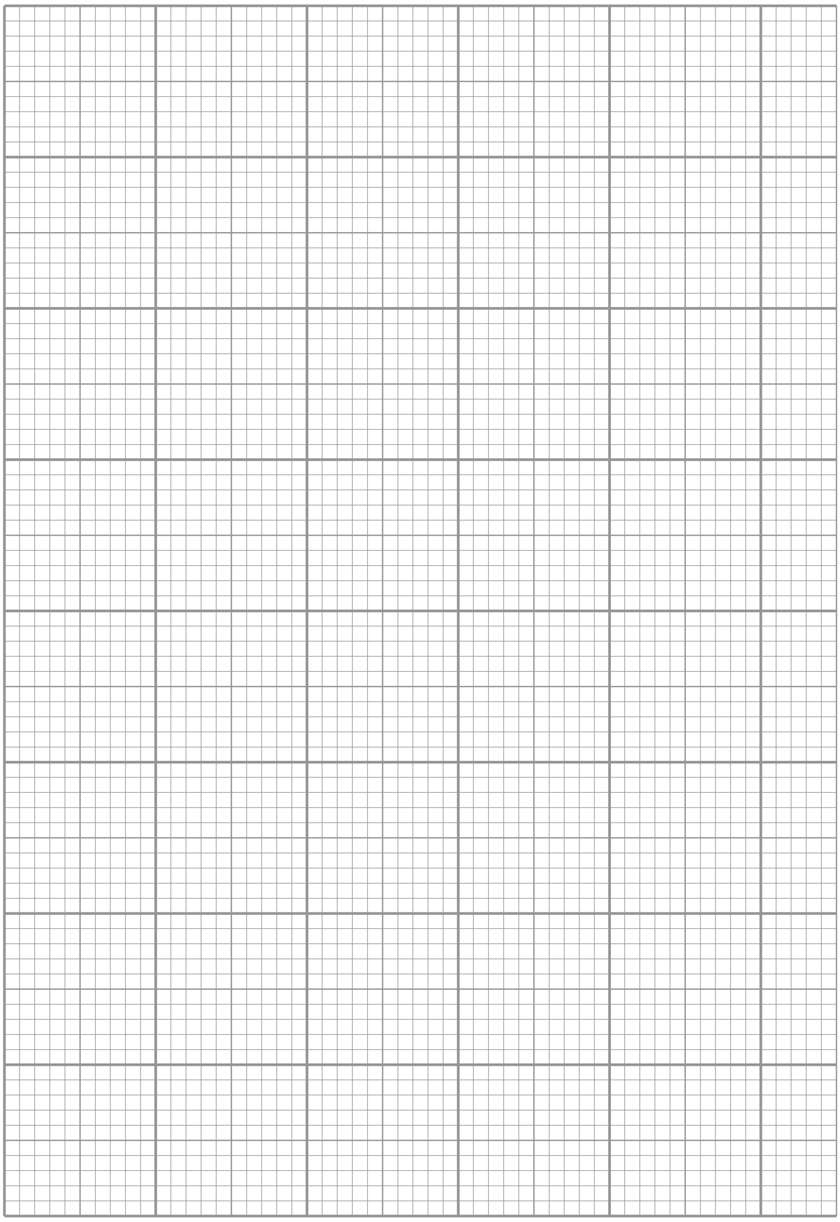
(b) An experiment is then carried out to find out how the acceleration of the marble depends on the height h_1. Throughout this experiment height h_2 is constant. The results of the experiment are shown in the table below.

Height h_1 (m)	Acceleration (m/s^2)
0	0
0.2	0.14
0.5	0.36
0.6	0.43
0.9	0.62
1.0	0.74

(i) Use the data in the table to plot a graph on the grid below and draw a suitable line. [3]



Examiner only

(ii) Use the graph to find the height required to produce an acceleration of $0.5\,m/s^2$. [1]

Height = ...

(iii) Describe how the experiment can be extended to determine how the range depends on the height h_2. [2]

....

....

....

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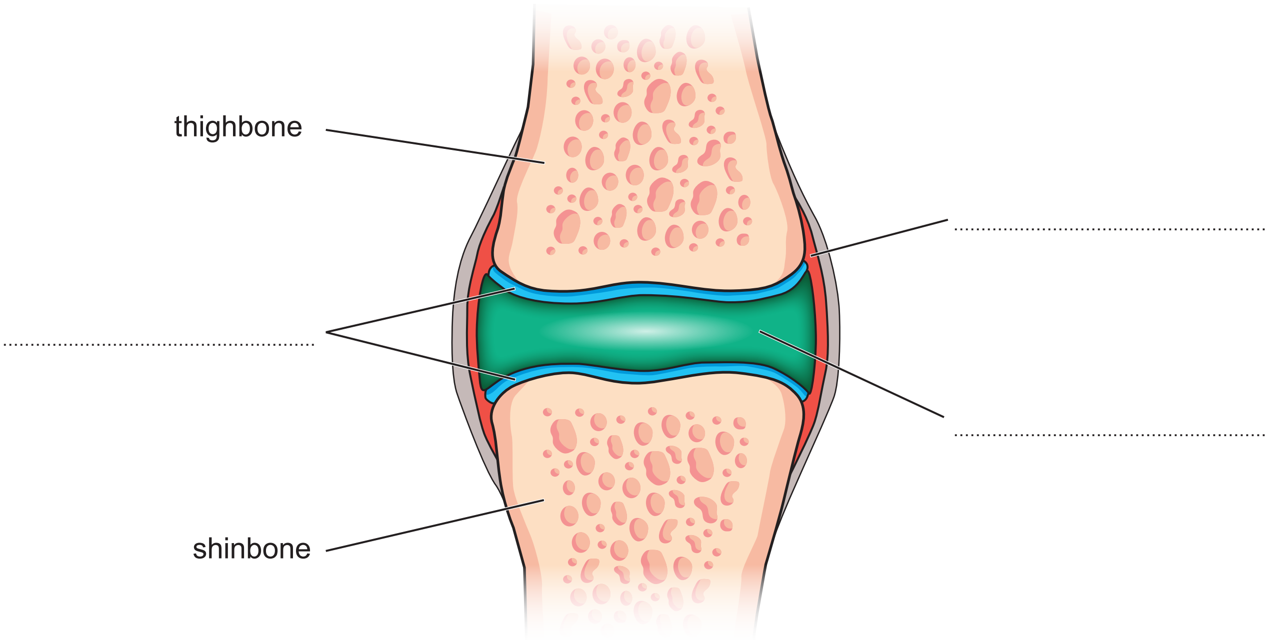
Examiner only

(c) Ski jumpers place their knee joints under considerable strain during their career. The knee is easily injured.

A simplified diagram of a knee joint is shown below.



(i) **Complete** the labelling of the diagram above. [3]

(ii) Explain why a ski jumper will not be able to compete if they damage the cartilage in their knee joints. [2]

....

....

....

Examiner only

(iii) An X-ray was taken of a ski jumper's leg following an accident.

State the name of this **type** of fracture. .... [1]

16